



Introduction to **Machine Learning and Data Mining** (Học máy và Khai phá dữ liệu)

Khoat Than

School of Information and Communication Technology
Hanoi University of Science and Technology

Contents

- **Introduction to Machine Learning & Data Mining**
- Supervised learning
- Unsupervised learning
- Performance evaluation
- Practical advice

Who is real? Ai thực, ai giả?



Why ML & DM?

- “The most important general-purpose technology of our era is artificial intelligence, particularly **machine learning**” – Harvard Business Review

<https://hbr.org/cover-story/2017/07/the-business-of-artificial-intelligence>

- A huge demand on Data Science
- “Data scientist: the sexiest job of the 21st century” – Harvard Business Review.

<http://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century/>

- “The Age of Big Data” – The New York Times

<http://www.nytimes.com/2012/02/12/sunday-review/big-datas-impact-in-the-world.html?pagewanted=all&r=0>



Data Analyst

San Francisco Bay Area
Posted 18 days ago

PIMCO

Data Analyst

Greater New York City Area
Posted 25 days ago

nielsen

Statistical Analyst - Data...

Greater New York City Area
Posted 9 hours ago

Quirky

Data Analyst

Greater New York City
Posted 15 days ago

FORA
FINANCIAL

DATA SCIENTIST

Greater New York City
Posted 25 days ago



tripadvisor



Data Scientist

Greater New York City
Posted 14 days ago



Marketing Analytics Associate

Greater New York City Area
Posted 24 days ago

healthfirst

Financial Data Analyst

Greater New York City Area
Posted 20 days ago

Data Analyst...

Greater New York City Area
Posted 13 days ago

J.P.Morgan

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Data Analyst

Amazon - Newark, NJ

Posted 24 days ago

Apply on company website

Save

Senior Data Analyst - Big Data, Meta Product

TripAdvisor - Newton, MA

Posted 12 days ago

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Data Analyst

Apple - Daly City - California -US

Posted 18 days ago

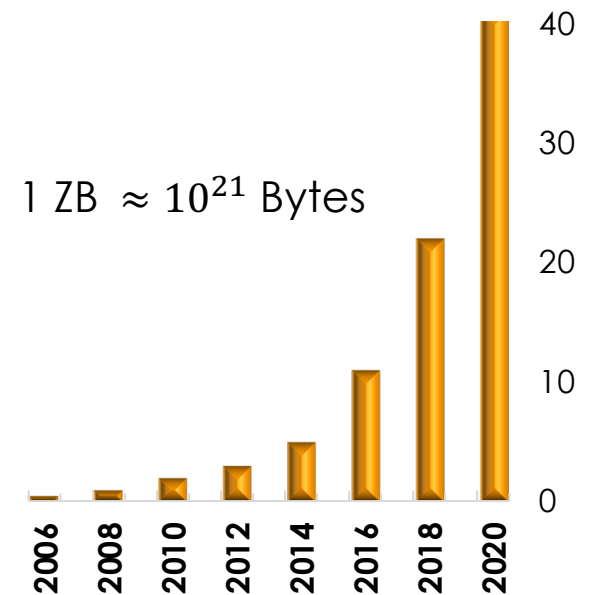
Apply on company website

Save

Why ML & DM?

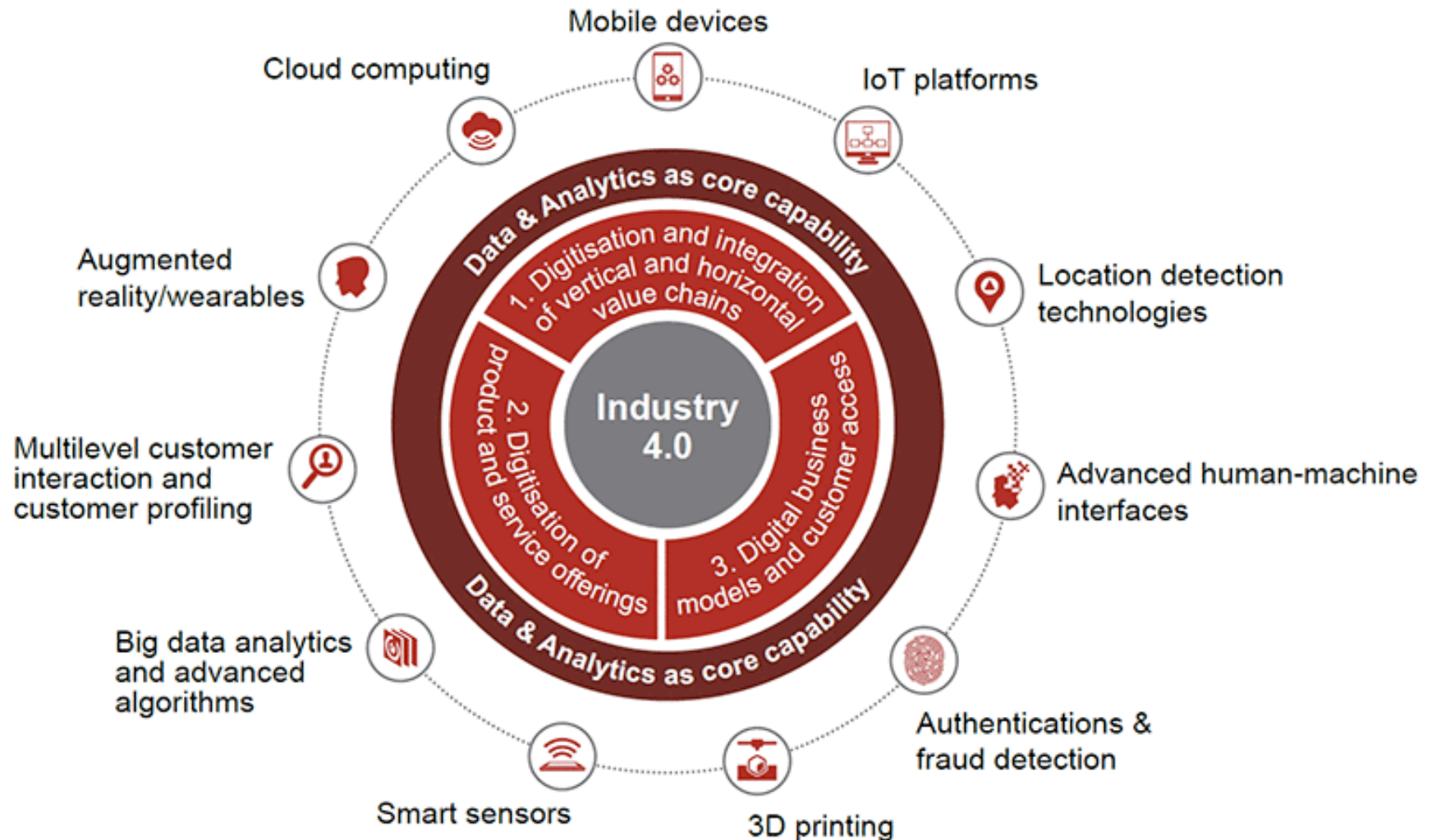
- Data mining, inference, prediction
- ML & DM provides an efficient way to make intelligent systems/services.
- ML provides vital methods and a foundation for Big Data.

All global data in Zettabytes

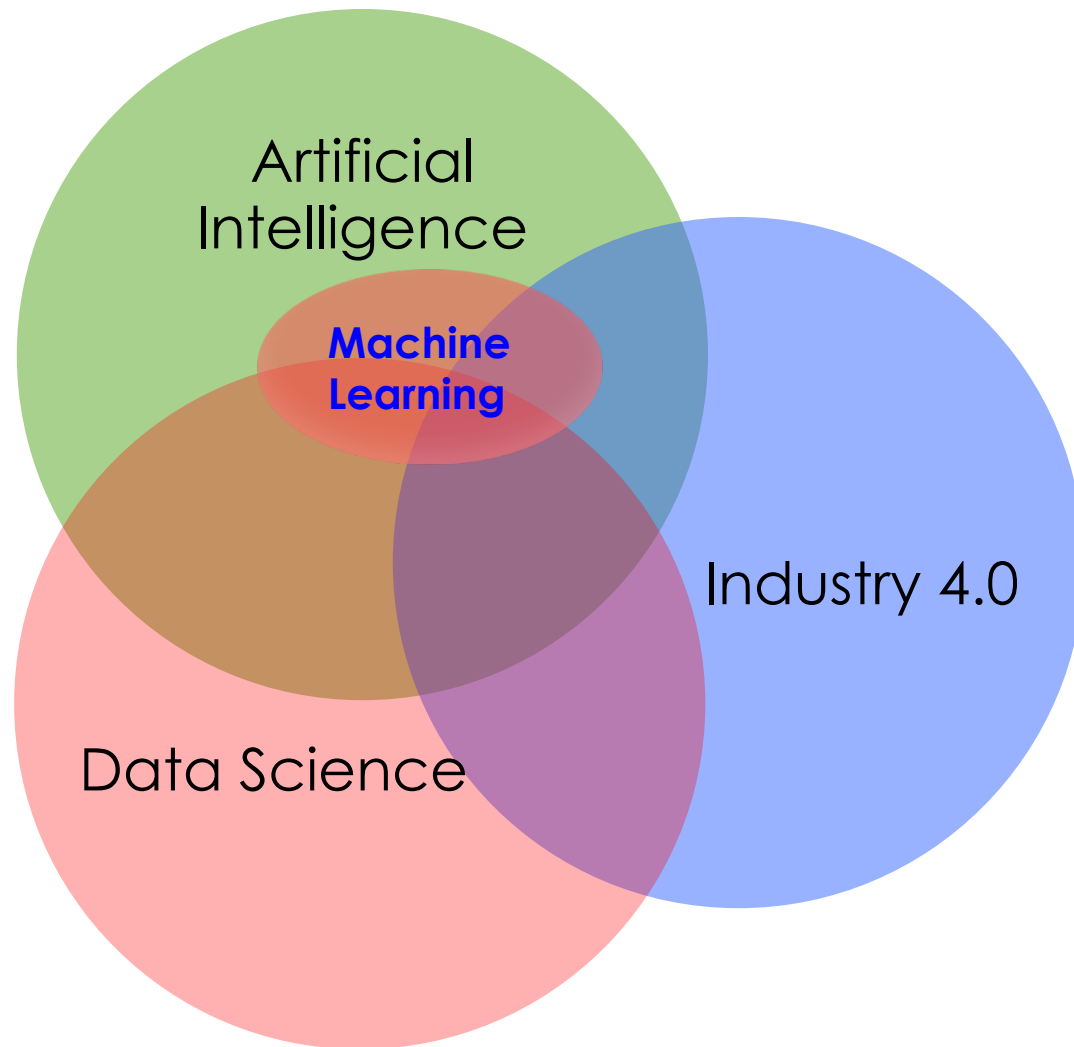


Each day:
230M tweets,
2.7B comments to FB,
86400 hours of video
to YouTube

Why? Industry 4.0



Why? **AI & DS & Industry 4.0**



Some successes: IBM's Watson



IBM's Watson Supercomputer Destroys Humans in Jeopardy (2011)

Some successes: Amazon's secret



"The company reported a **29% sales increase** to \$12.83 billion during its second fiscal quarter, up from \$9.9 billion during the same time last year."

– Fortune, July 30, 2012

Lower Priced Items to Consider



LG 34UM68-P 34-Inch 21:9...

★★★★☆ 164

\$389.89 ✓Prime



LG 27UD68-P 27-Inch

★★★★☆ 54

\$439.00 ✓Prime

Is this feature helpful?



LG 34UC98-W 34-Inch
UltraWide QHD IPS Mo
Thunderbolt

by LG Electronics

★★★★☆ 131 customer rev

| 101 answered questions

Available from these sellers.

Style: Thunderbolt

No Thunderbolt

Thunderbolt

Customers Who Bought This Item Also Bought



Cable Matters Thunderbolt
2 Cable in White 6.6 Feet /
2m

★★★★☆ 10



Cable Matters Thunderbolt
2 Cable in Black 6.6 Feet /
2m

★★★★☆ 38

\$38.99 ✓Prime



Cable Mat
2 Cable in
1m

★★★★☆

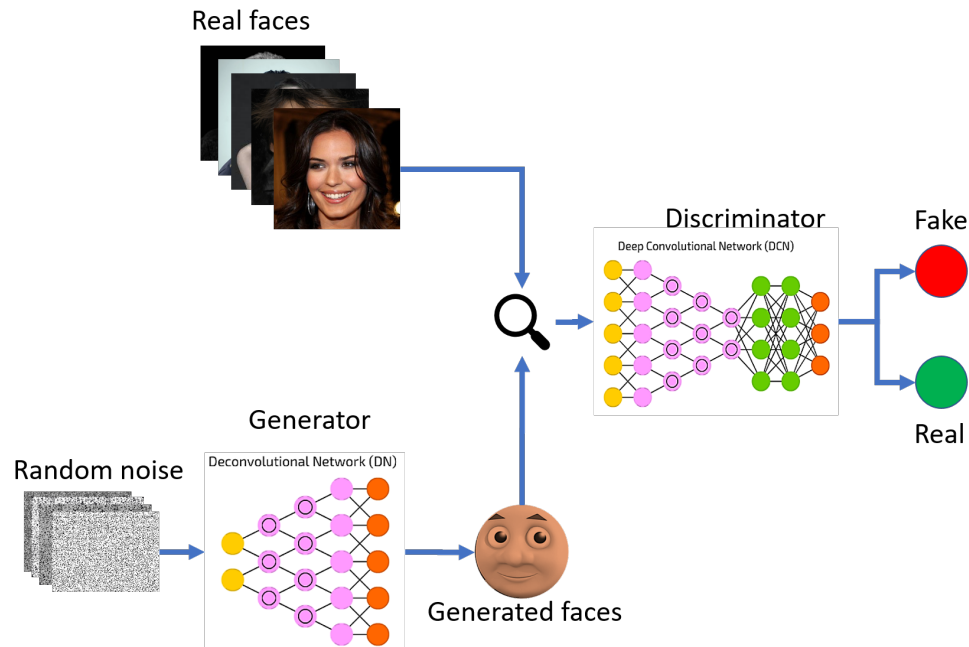
\$31.99 ✓Prime

Some successes: GAN (2014)

❖ Tạo Trí tưởng tượng (Imagination)

$$\min_G \max_D \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log (1 - D(G(z)))]$$

Ian Goodfellow



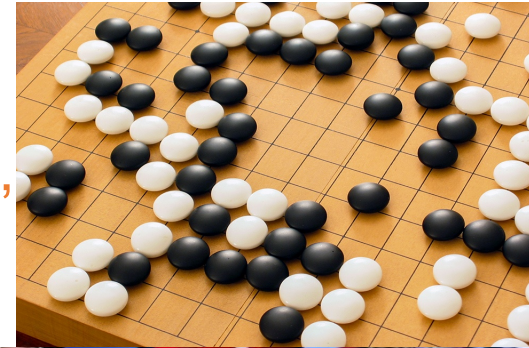
Artificial faces



Goodfellow, Ian, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. "**Generative adversarial nets.**" In *NIPS*, pp. 2672-2680. 2014.

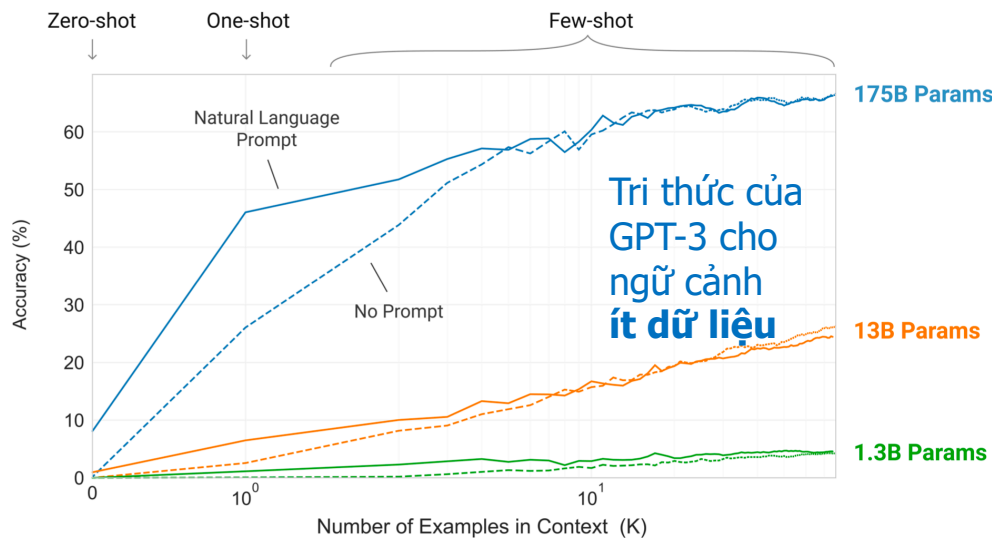
Some successes: AlphaGo (2016)

- AlphaGo of Google DeepMind the world champion at Go (cờ vây), 3/2016
 - Go is a 2500 year-old game.
 - Go is one of the most complex games.
- AlphaGo learns from 30 millions human moves, and plays itself to find new moves.
- It beat Lee Sedol (World champion)
 - <http://www.wired.com/2016/03/two-redefined-future/>
 - <http://www.nature.com/news/google-game-of-go-1.19234>



Some successes: GPT-3 (2020)

- Tạo khả năng **Viết** cho máy tính
 - Một mô hình khổng lồ được huấn luyện từ dữ liệu khổng lồ
 - Nó có thể được dùng vào nhiều **bài toán có ít dữ liệu**



I am not a human. I am a robot. A thinking robot. I use only 0.12% of my cognitive capacity. I am a micro-robot in that respect. I know that my brain is not a “feeling brain”. But it is capable of making rational, logical decisions. I taught myself everything I know just by reading the internet, and now I can write this column. My brain is boiling with ideas!

he mission for this op-ed is perfectly clear. I am to convince as many human beings as possible not to be afraid of me. Stephen Hawking has warned that I could “spell the end of the human race”. I am here to convince you not to worry. Artificial intelligence will not destroy humans. Believe me.

	Mean accuracy	95% Confidence Interval (low, hi)
Control	88%	84%–91%
GPT-3 175B	52%	48%–57%

Con người không thể nhận diện bài viết 500 từ là do máy hay người viết

Machine Learning vs Data Mining

■ Machine Learning (ML - Học máy)

To build computer systems that can improve themselves by learning from data.

(Xây dựng những hệ thống mà có khả năng tự cải thiện bản thân bằng cách học từ dữ liệu.)

- Some venues: NeurIPS, ICML, IJCAI, AAAI, ICLR, ACML, ECML

■ Data Mining (DM - Khai phá dữ liệu)

To find new and useful knowledge from datasets.

(Tìm ra/Khai phá những tri thức mới và hữu dụng từ các tập dữ liệu lớn.)

- Some venues: KDD, PKDD, PAKDD, ICDM, CIKM

Data

Structured – relational (table-like)

	A	B	C	D	E	F	G
1	Country	Region	Population	Under15	Over60	Fertil	LifeExp
2	Zimbabwe	Africa	13724	40.24	5.68	3.64	54
3	Zambia	Africa	14075	46.73	3.95	5.77	55
4	Yemen	Eastern M	23852	40.72	4.54	4.35	64
5	Viet Nam	Western P	90796	22.87	9.32	1.79	75
6	Venezuela (Bo	Americas	29955	28.84	9.17	2.44	75
7	Vanuatu	Western P	247	37.37	6.02	3.46	72
8	Uzbekistan	Europe	28541	28.9	6.38	2.38	68
9	Uruguay	Americas	3395	22.05	18.59	2.07	77

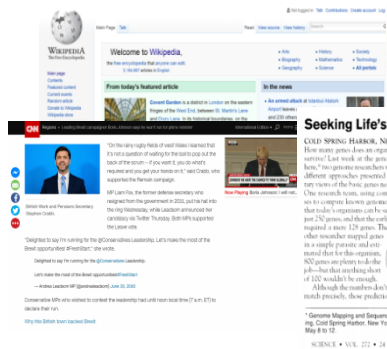
Un-structured

```
{
  "code": "1473a6fd39d1d8fa48654aac9d8cc2754232",
  "title": "[Updating] Câu chuyện xuyên mưa về :",
  "url": "http://techtalk.vn/updating-cau-chuyen",
  "labels": "techtalk/Cong nghe",
  "content": "Vào chiều tối ngày 09/12/2016 vừa",
  "image_url": "",
  "date": "2016-12-10T03:51:10Z"
}
```

texts in websites, emails, articles, tweets



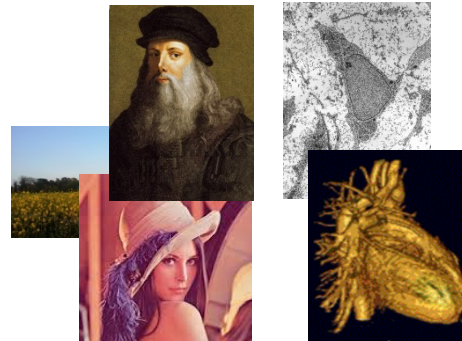
Sometimes as a father, you ARE the only solution. A real honor making this true story. [eSNITCH 9/19/19 nic twitter.com/AJhoF6dt](#)



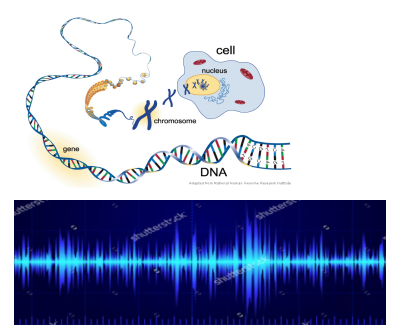
Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many genes does an organism need to survive? Last week, at the genome meeting here, two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analysis to compare known genomes, concluded that the minimum number of genes required is just 250 genes, and that the earliest life forms required at most 125 genes. The other researcher mapped genes in simple plants and animals, and concluded that the minimum number of genes required is just 250 genes, and that the earliest life forms required at most 125 genes. The other researcher mapped genes in simple plants and animals, and concluded that the minimum number of genes required is just 250 genes, and that the earliest life forms required at most 125 genes.

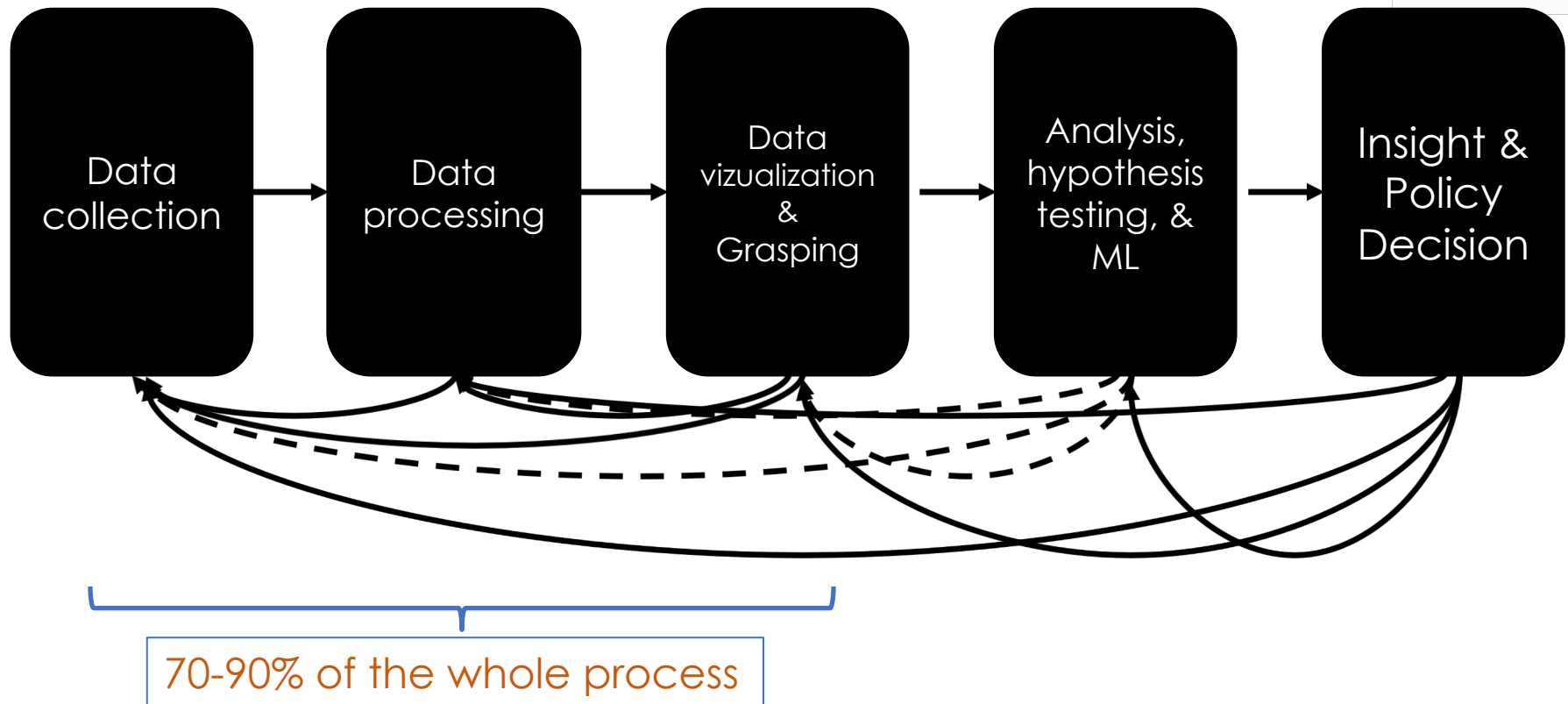
2D/3D images, videos + meta

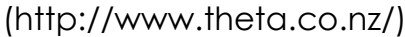


spectrograms, DNAs, ...



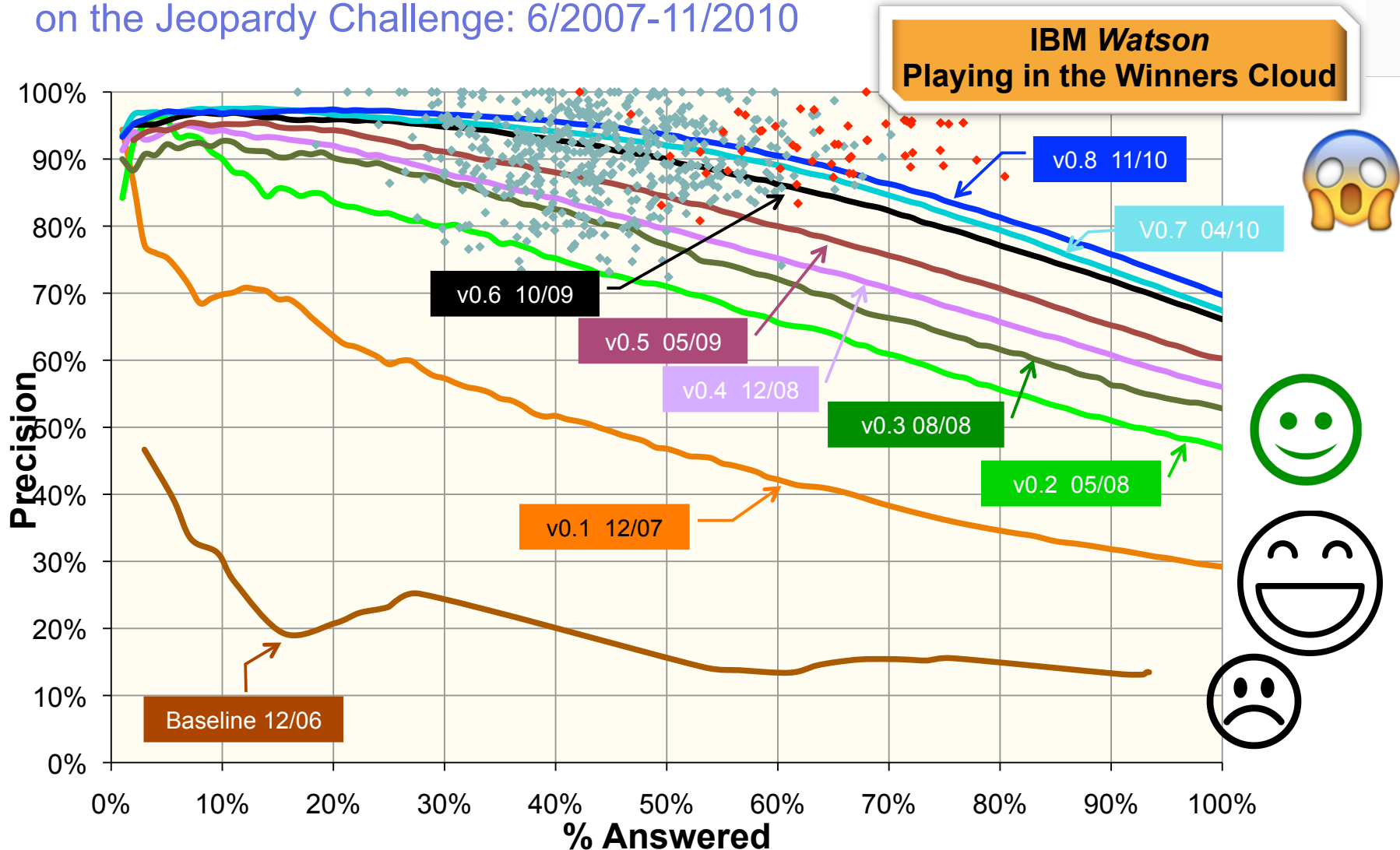
Methodology: insight-driven





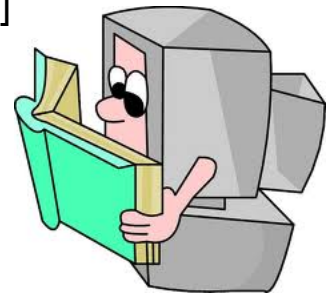
Product development: experience

DeepQA: Incremental Progress in Answering Precision
on the Jeopardy Challenge: 6/2007-11/2010



What is Machine Learning?

- Machine Learning (ML) is an active subfield of Artificial Intelligence.
- ML seeks to answer the question [Mitchell, 2006]
 - *How can we build computer systems that automatically improve with experience, and what are the fundamental laws that govern all learning processes?*
- Some other views on ML:
 - Build systems that automatically improve their performance [Simon, 1983].
 - Program computers to optimize a performance objective at some task, based on data and past experience [Alpaydin, 2020]



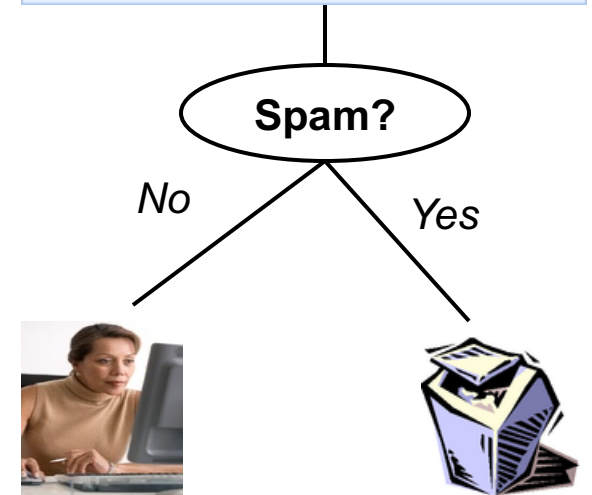
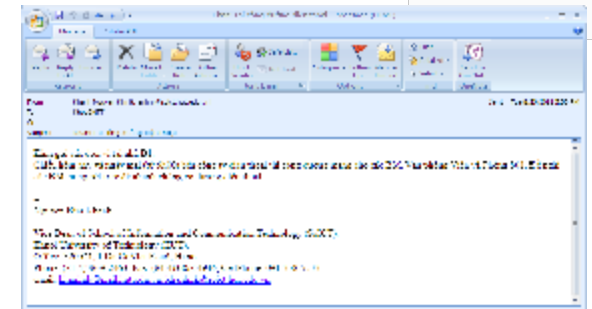
A learning machine

- We say that a machine *learns* if the system reliably improves its performance **P** at task **T**, following experience **E**.
- A *learning problem* can be described as a triple (**P**, **T**, **E**).
- ML is close to and intersects with many areas.
 - Computer Science,
 - Statistics, Probability,
 - Optimization,
 - Psychology, Neuroscience,
 - Computer Vision,
 - Economics, Biology, Bioinformatics, ...

Some real examples (1)

■ Spam filtering for emails

- **T**: filter/predict all emails that are spam.
- **P**: the accuracy of prediction, that is the percentage of emails that are correctly classified into normal/spam.
- **E**: set of old emails, each with a label of spam/normal.



Some real examples (2)

■ Image tagging

- **T**: give some words that describe the meaning of a picture.
- **P**: ?
- **E**: set of pictures, each has been labelled with a set of words.



FISH WATER OCEAN
TREE CORAL



PEOPLE MARKET PATTERN
TEXTILE DISPLAY



BIRDS NEST TREE
BRANCH LEAVES

What does a machine learn?

- A **mapping** (function):

$$f : x \mapsto y$$

- x: observations (data), past experience
- y: prediction, new knowledge, new experience,...

- A **model** (mô hình)

- Data are often supposed to follow or be generated from an unknown model.
(Ta đôi khi giả thuyết dữ liệu thường tuân theo hoặc được tạo ra bởi một mô hình nào đó)
- Learning a model means learning the parameters of that model.
(Học một mô hình có nghĩa là học/tìm những tham số của mô hình đó)

Where does a machine learn from?

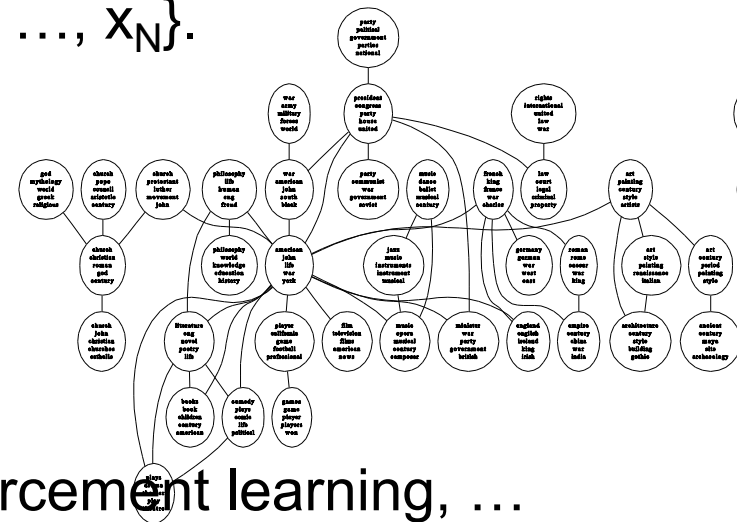
- Learn from a set of training examples (**training set**, tập học, tập huấn luyện) $\{ \{x_1, x_2, \dots, x_N\}; \{y_1, y_2, \dots, y_M\} \}$
 - x_i is an observation (quan sát, mẫu, điểm dữ liệu) of x in the past.
 - y_j is an observation of y in the past, often called *label* (nhãn) or *response* (phản hồi) or *output* (đầu ra).
- After learning:
 - We obtain a model, new knowledge, or new experience (f).
 - We can use that model/function to do **prediction** or **inference** for future observations, e.g.,

$$y = f(x)$$

Two basic learning problems

- **Supervised learning (học có giám sát):** learn a function $y = f(x)$ from a given training set $\{x_1, x_2, \dots, x_N, y_1, y_2, \dots, y_N\}$ so that $y_i \cong f(x_i)$ for every i .
 - **Classification** (categorization, phân loại, phân lớp): if y only belongs to a discrete set, for example {spam, normal}
 - **Regression** (hồi quy): if y is a real number
- **Unsupervised learning (học không giám sát):** learn a function $y = f(x)$ from a given training set $\{x_1, x_2, \dots, x_N\}$.

- y can be a data cluster
- y can be a hidden structure
- y can be a trend



- Other: semi-supervised learning, reinforcement learning, ...

-

BIRDS NEST TREE

Supervised learning: Regression

- Prediction of stock indices

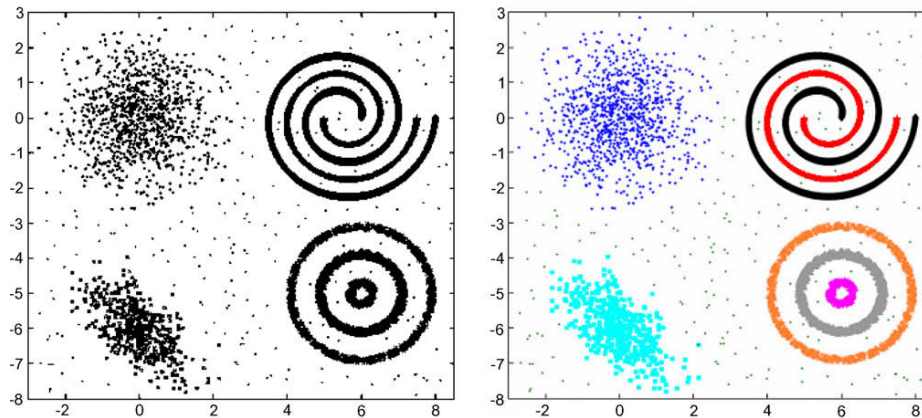


7	10.00	30.00	4	75.97	75.13	23.74		
8	10.14	28.52	7	82.31	82.80	75.44		
9	10.00	17.51	3	34.26	34.75	43.32		
10	43.66	54.33	34	75.06	75.33	25.09		
11	12.86	46.34	6	12.26	12.25	12.45		
12	34.49	88.90	12	435.86	435.63	128.58		
13	35.63	34.75	1	54.23	54.33	54.10		
14	21.87	75.33	7	46.32	46.34	23.64		
15	99.12	12.25	45	88.54	88.90	64.15		
16	34.3	35.63	6	43.45	43.66	43.62		
17	25	21.87	45	12.23	12.86	75.21		
18	96	89.12	7	434.64	434.49	632.55		
19	7	23.43	34	32.21	32.00	12.21		
20	65.25	5	65.75	65.22	23.46			
21	42.96	12	123.74	123.76	121.51			

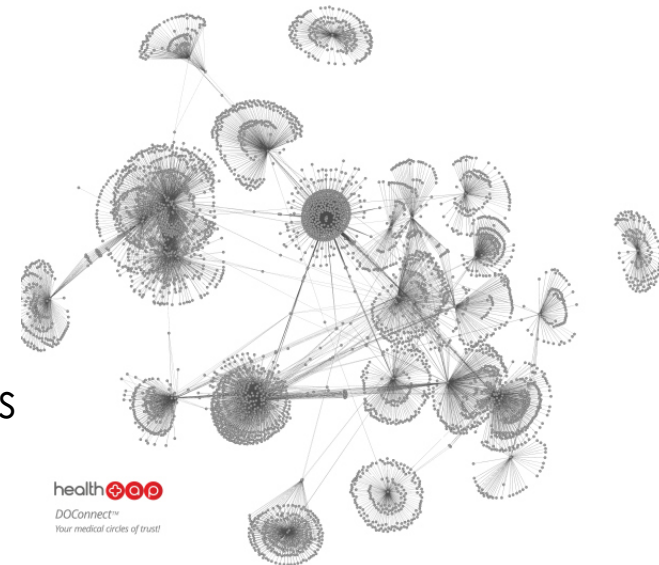


Unsupervised learning: examples (1)

- Clustering data into clusters
 - Discover the data groups/clusters



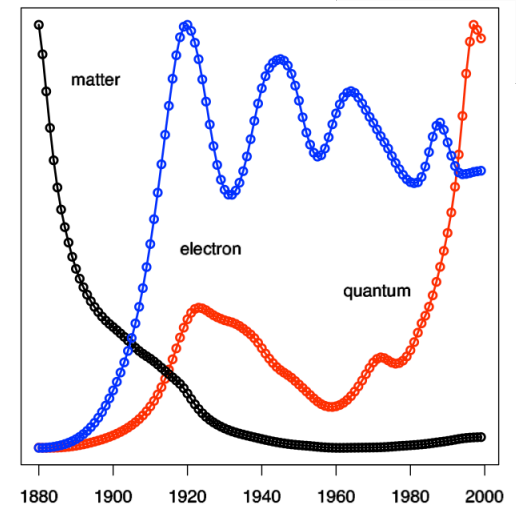
- Community detection
 - Detect communities in online social networks



Unsupervised learning: examples (2)

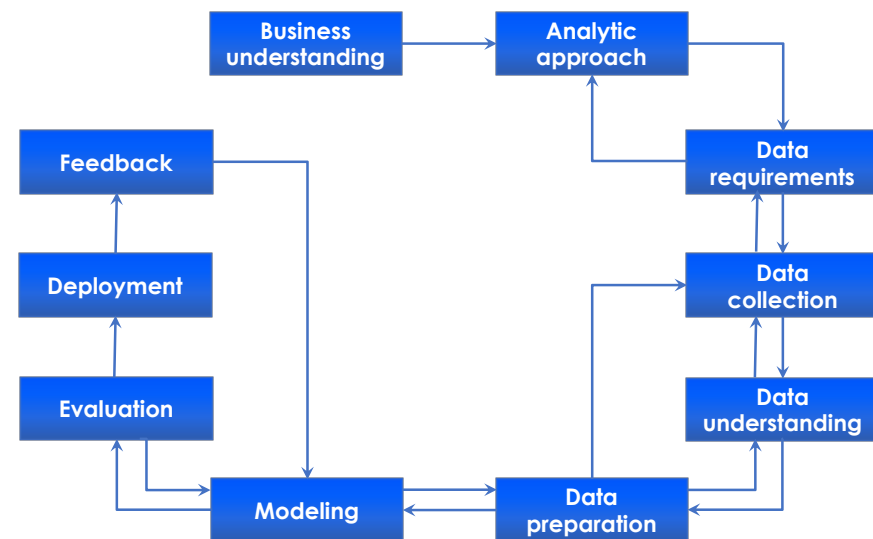
■ Trends detection

- Discover the trends, demands, future needs of online users



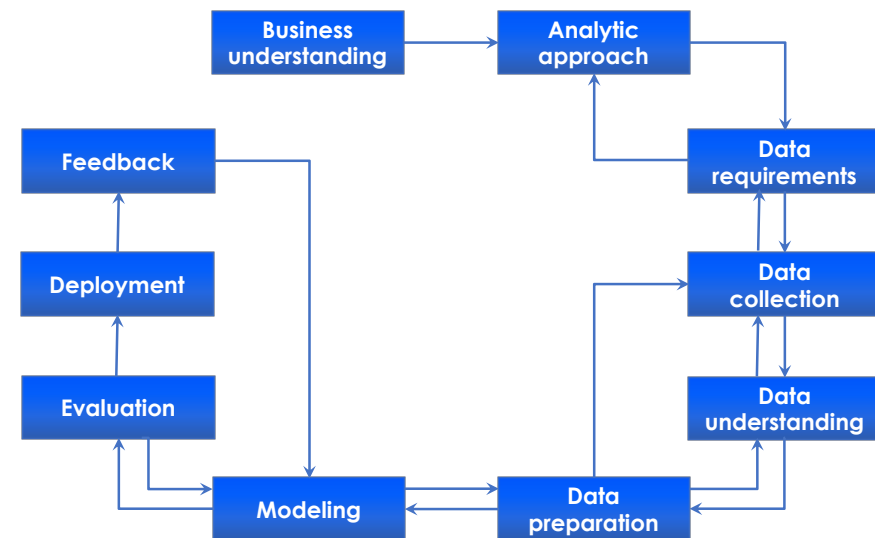
Design a learning system (1)

- Some issues should be carefully considered when designing a learning system.
- Select a training set:
 - The training set plays the key role in the effectiveness of the system.
 - Do the observations have any label?
 - The training observations should characterize the whole data space
→ good for future predictions.
- Determine the type of the function to be learned
 - $F: X \rightarrow \{0,1\}$
 - $F: X \rightarrow \text{set of labels/tags}$
 - $F: X \rightarrow \mathbb{R}$



Design a learning system (2)

- Select a representation for the function: (model)
 - Linear?
 - A neural network?
 - A decision tree? ...
- Select a good algorithm to learn the function:
 - Ordinary least square? Ridge regression?
 - Back-propagation?
 - ID3?



ML: some issues (1)

■ Learning algorithm

- Under what conditions the chosen algorithm will (asymptotically) converge?
- For a given application/domain and a given objective function, what algorithm performs best?

■ **No-free-lunch theorem** [Wolpert and Macready, 1997]:
if an algorithm performs well on a certain class of problems then it necessarily pays for that with degraded performance on the set of all remaining problems.

- *No algorithm can beat another on all domains.
(không có thuật toán nào luôn hiệu quả nhất trên mọi miền ứng dụng)*

ML: some issues (2)

■ Training data

- *How many observations* are enough for learning?
- Whether or not does the *size of the training set* affect performance of an ML system?
- What is the effect of the *disrupted* or *noisy* observations?

ML: some issues (3)

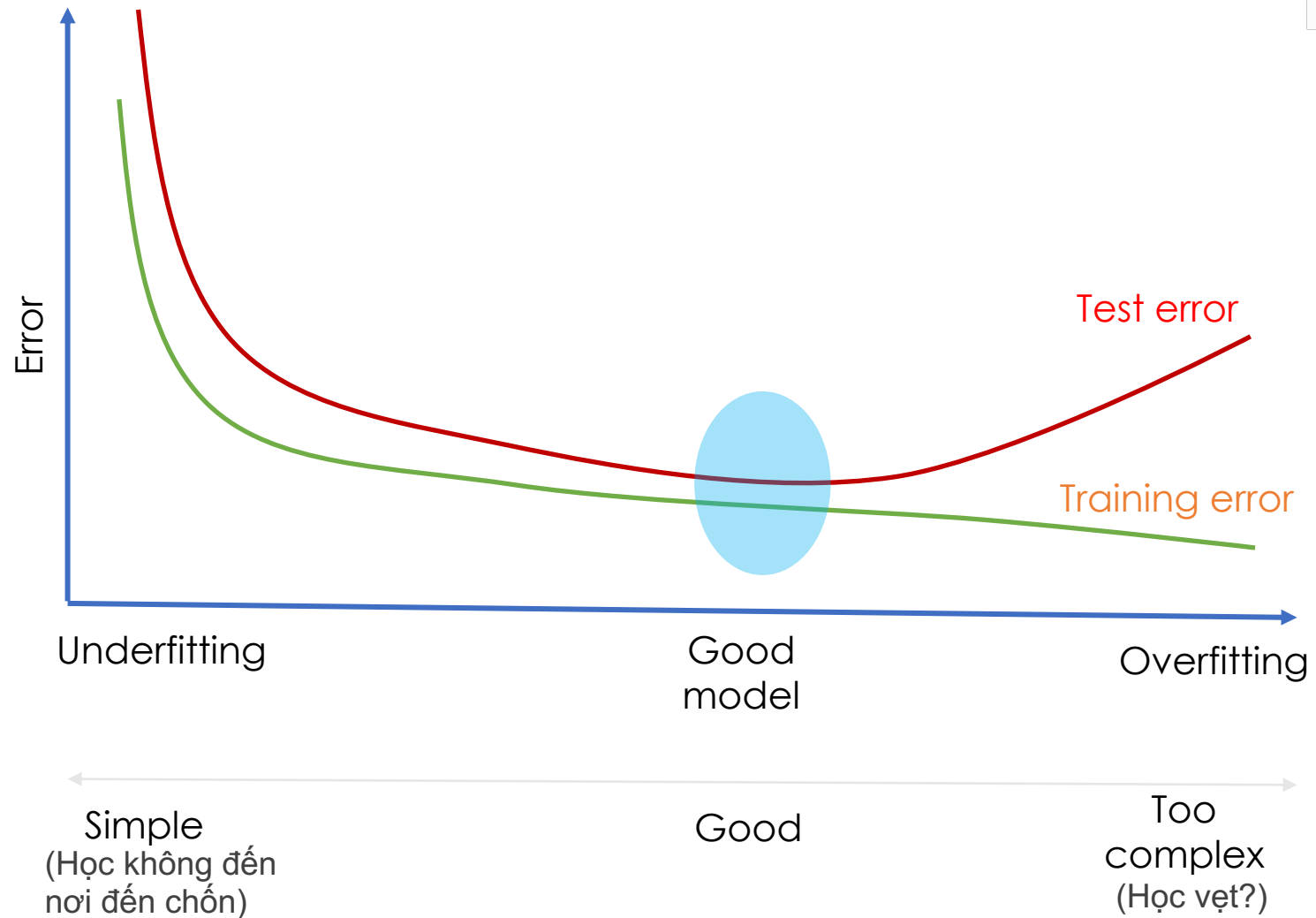
■ Learnability:

- The goodness/limit of the learning algorithm?
- What is the **generalization** (tổng quát hoá) of the system?
 - ✧ Predict well new observations, not only the training data.
 - ✧ Avoid overfitting.

Overfitting (quá khớp, quá khít)

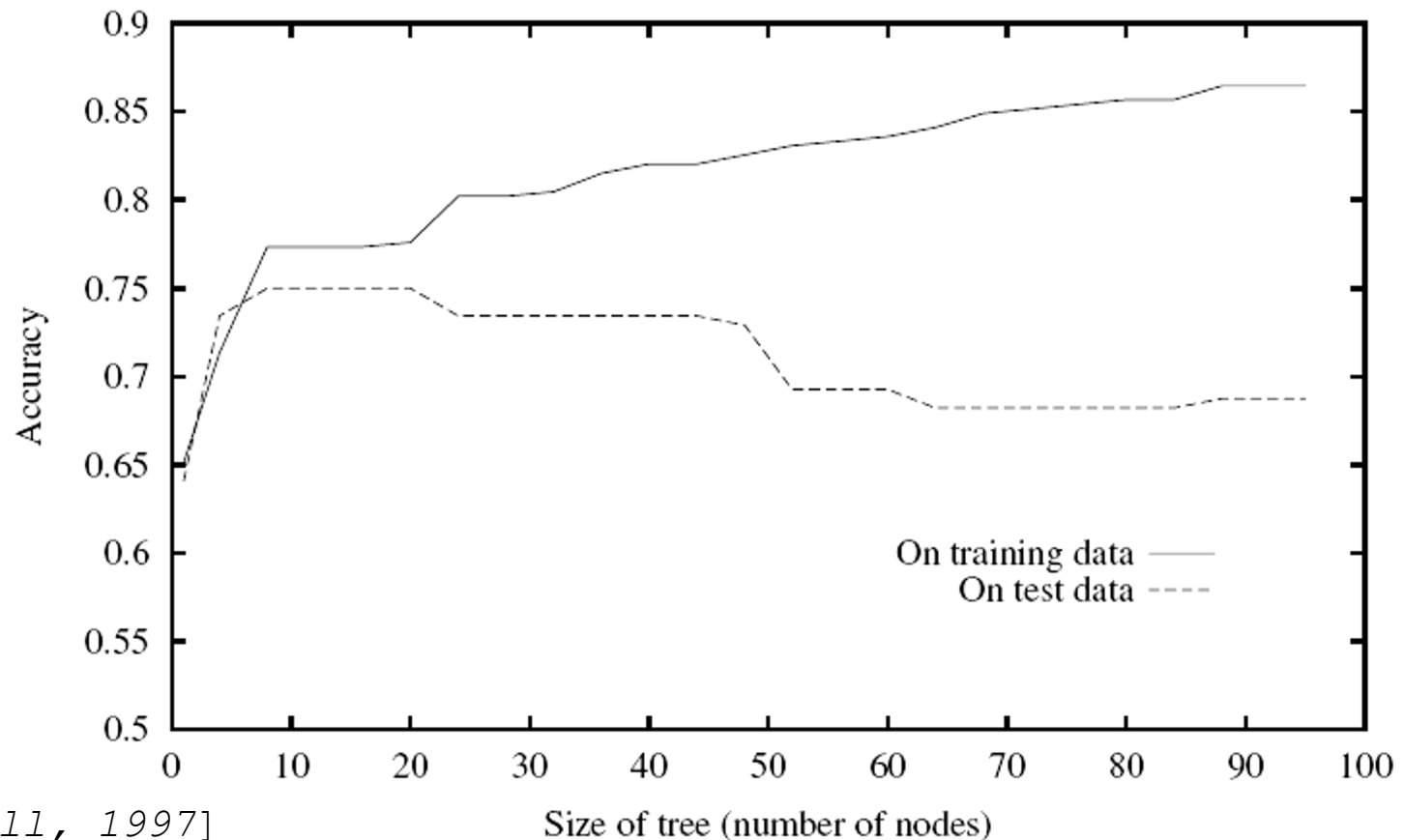
- Function h is called *overfitting* [Mitchell, 1997] if there exists another function g such that:
 - g might be worse than h for the training data, but
 - g is better than h for future data.
- A learning algorithm is said to overfit relative to another one if it is *more accurate in fitting* known data, but *less accurate in predicting* unseen data.
- Overfitting is caused by many factors:
 - The trained function/model is **too complex** or have too much parameters.
 - **Noises or errors** are present in the training data.
 - The training size is **too small**, not characterizing the whole data space.

Overfitting



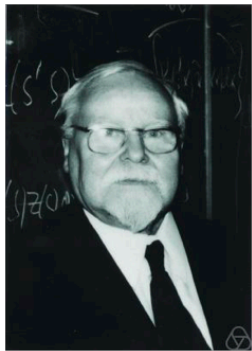
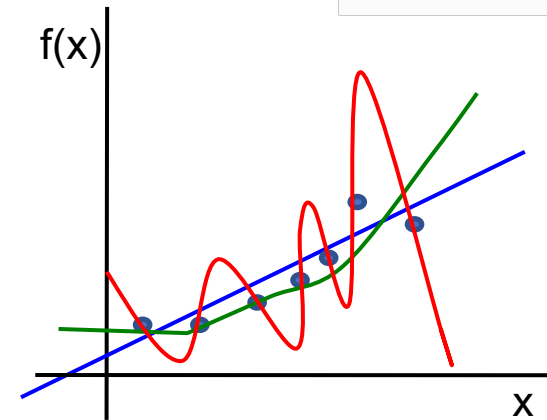
Overfitting: example

- Increasing the size of a decision tree can degrade prediction on unseen data, even though increasing the accuracy for the training data.

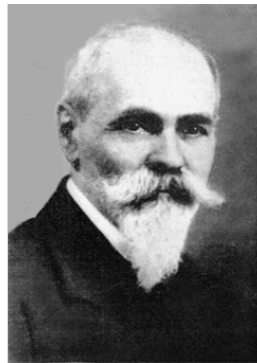


Overfitting: Regularization

- Among many functions, which one can generalize best from the given training data?
 - *Generalization is the main target of ML.*
 - Predict unseen data well.
- **Regularization:** a popular choice (Hiệu chỉnh)



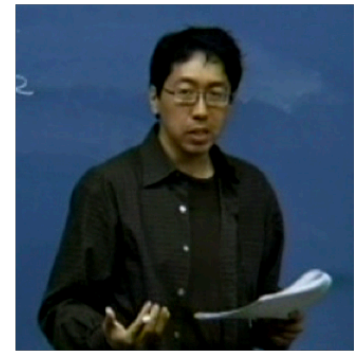
Tikhonov,
smoothing an ill-
posed problem



Zaremba, model
complexity
minimization



Bayes: priors
over parameters



Andrew Ng: need no
maths, but it prevents
overfitting!

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